Nuclear Astrophysics

Advanced Burning Stages: Carbon, Neon, Oxygen, Silicon
To the Iron Peak

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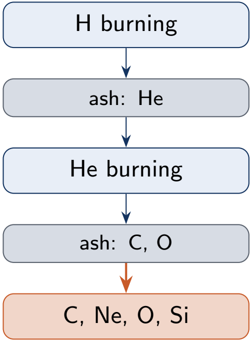
Where we left off

- Hydrogen burning (p–p chain, CNO) $\rightarrow$ helium ash.
- Helium burning (triple-α) $\rightarrow$ $^{12}\mathrm{C}$ and $^{16}\mathrm{O}$; further α-capture is *blocked* at $^{16}\mathrm{O}$ by $^{20}\mathrm{Ne}$ selection rules.
- Result of helium burning: a core dominated by $^{12}\mathrm{C}$ and $^{16}\mathrm{O}$, with C/O $\approx$ 0.6.



Today's territory

Four advanced burning stages above helium: C, Ne, O, Si. Below the iron peak only — neutron-induced nucleosynthesis comes next lecture.

The contraction–ignition cycle

After each fuel is exhausted, the same self-correcting mechanism repeats:

1. Fuel runs out $\Rightarrow$ thermal pressure drops, gravity wins.
2. Core **contracts**, ρ and T both rise.
3. Once T crosses the next Coulomb barrier, the ashes ignite and a new burning stage begins.
4. Energy production resumes; hydrostatic balance is restored *until* this fuel is exhausted too.

Two possible endings

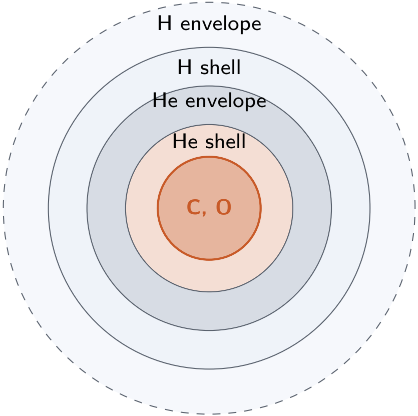
- Insufficient M to ignite the next fuel $\Rightarrow$ electron-degenerate white dwarf.
- Massive enough to traverse all stages $\Rightarrow$ ends in a **supernova** explosion.

Why timescales shrink

Every stage past helium suffers heavy neutrino losses. Energy that should support the star is radiated away $\Rightarrow$ the star must burn faster, and the next stage lasts a fraction of the previous one.

The onion structure of an evolved massive star

Burning of earlier fuels continues in shells around the contracting core — the star develops a layered, onion-like geometry.



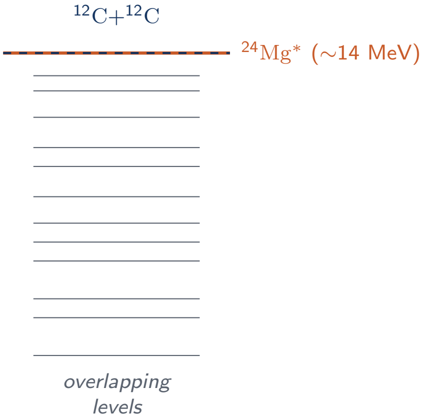
- Core: ashes of helium burning $\Rightarrow$ $^{12}\mathrm{C}$ and $^{16}\mathrm{O}$.
- Surrounding shells continue to burn helium (then hydrogen) — they are *not* "gone" just because the core moved on.
- Each successive advanced burning stage adds a new composition layer underneath, while the outer shells keep burning.

Reading the onion

Outer shells $\rightarrow$ lighter fuels (younger material).
Inner shells $\rightarrow$ heavier ash (more recent burning).

Carbon burning — the first heavy-nucleus fusion

- Core composition after He burning: $^{12}\mathrm{C}$ and $^{16}\mathrm{O}$.
- Coulomb barrier favours $^{12}\mathrm{C}+^{12}\mathrm{C}$ over O+O or C+O $\Rightarrow$ **carbon ignites first**.
- This is the *first* fusion between heavy ($A \geq 12$) nuclei in stellar evolution.
- Compound nucleus: $^{24}\mathrm{Mg}^{*}$ with $\Delta m \approx 14\,\mathrm{MeV}$ excitation $\Rightarrow$ a dense forest of overlapping levels; reaction proceeds statistically through them.



Five exit channels of $^{12}\mathrm{C}+^{12}\mathrm{C}$

Reaction	Q-value	Comment
$^{12}\mathrm{C}(^{12}\mathrm{C}, p)^{23}\mathrm{Na}$	2241 keV	exothermic, dominant
$^{12}\mathrm{C}(^{12}\mathrm{C}, \alpha)^{20}\mathrm{Ne}$	4617 keV	exothermic, dominant
$^{12}\mathrm{C}(^{12}\mathrm{C}, n)^{23}\mathrm{Mg}$	-2599 keV	endothermic (n source)
$^{12}\mathrm{C}(^{12}\mathrm{C}, \gamma)^{24}\mathrm{Mg}$	–	radiative; minor
$^{12}\mathrm{C}(^{12}\mathrm{C}, ^{8}\mathrm{Be})^{16}\mathrm{O}$	–	minor; $^{8}\mathrm{Be}$ unstable

- The three *light-particle* exits (p, α, n) dominate the energy budget.
- The last two are weak channels but contribute to **element synthesis** (e.g. a route back to $^{16}\mathrm{O}$).
- Neutron channel is endothermic — only matters at higher T, but its neutrons feed later neutron-capture chains.

When does carbon ignite? The Chandrasekhar limit

Temperatures (in GK)

- Quiescent C burning: $T_9 = 0.6\text{–}1.0$
- Explosive C burning: $T_9 = 1.8\text{–}2.5$

- Ignition requires both high T *and* sufficient mass to *reach* that T during contraction.
- Chandrasekhar limit: $M_{\rm Ch} = 1.4\,M_\odot \approx 2.765 \times 10^{30}$ kg.

Two fates of a C/O core

- $M < M_{\rm Ch}$: core never gets hot enough. Electron degeneracy pressure halts contraction $\Rightarrow$ white dwarf, eventually a **black dwarf** as it cools.
- $M > 8\text{–}10\,M_\odot$: contracting core remains non-degenerate; at $\rho \sim 3 \times 10^6$ g/cm^3 and $T_9 \approx 0.6$, carbon ignites.

Why neon before oxygen?

At the end of core C burning, the most abundant ash nuclei are $^{16}\mathrm{O}$, $^{20}\mathrm{Ne}$, $^{23}\mathrm{Na}$, and $^{24}\mathrm{Mg}$.

- $^{16}\mathrm{O}$, $^{23}\mathrm{Na}$, $^{24}\mathrm{Mg}$ are inert against photodisintegration even at the rising T.
- $^{20}\mathrm{Ne}$ is the exception: its α-separation energy is only $S_\alpha = 4.73$ **MeV**, the lowest of the group.
- So as T climbs, γ's in the Planck tail start to photo-disintegrate $^{20}\mathrm{Ne}$ before O+O can ignite.
- Although there is more $^{16}\mathrm{O}$ than $^{20}\mathrm{Ne}$, S_α – not abundance – decides the next stage.

α-separation energies (MeV)

Nucleus	S_α
$^{20}\mathrm{Ne}$	4.73
$^{16}\mathrm{O}$	7.16
$^{23}\mathrm{Na}$	10.09
$^{24}\mathrm{Mg}$	9.32

The reactions that make up neon burning

Trigger reaction (endothermic photodisintegration):

$$^{20}\mathrm{Ne}(\gamma,\alpha)^{16}\mathrm{O}, \qquad Q = -4730\,\mathrm{keV}$$

Direct measurement is hard; use reciprocity theorem on the reverse reaction $^{16}\mathrm{O}(\alpha,\gamma)^{20}\mathrm{Ne}$, which is easy to do in the lab with an α beam. Result: $\lambda_\gamma(^{20}\mathrm{Ne}) \approx 1.5 \times 10^{-6}\ \mathrm{s}^{-1}$.

Follow-up captures (liberated α feeds the network)

$^{20}\mathrm{Ne}(\alpha,\gamma)^{24}\mathrm{Mg}(\alpha,\gamma)^{28}\mathrm{Si}$ $\quad (Q_{^{20}\mathrm{Ne}(\alpha,\gamma)} = 9316$ keV; $Q_{^{24}\mathrm{Mg}(\alpha,\gamma)} = 9984$ keV$)$

$^{23}\mathrm{Na}(\alpha,p)^{26}\mathrm{Mg}(\alpha,n)^{29}\mathrm{Si}$ $\quad (Q_{^{23}\mathrm{Na}(\alpha,p)} = 1821$ keV; $Q_{^{26}\mathrm{Mg}(\alpha,n)} = 34$ keV$)$

Net effect

Net consumption of 2 $^{20}\mathrm{Ne}$ to make $^{16}\mathrm{O} + ^{24}\mathrm{Mg}$, with follow-up captures building $^{28}\mathrm{Si}$. Temperatures: $T_9 = 1.2$–1.8 (quiescent), 2.5–3.0 (explosive).

Oxygen burning — the proliferation of exit channels

After neon burning the dominant nuclei are $^{16}\mathrm{O}$, $^{24}\mathrm{Mg}$, $^{28}\mathrm{Si}$ (Coulomb barriers $\approx$ 15, 21, 24 MeV) — oxygen has the lowest barrier, so it ignites next.

$$^{16}\mathrm{O} + ^{16}\mathrm{O} \rightarrow \, ^{32}\mathrm{S}^* \quad (\Delta m \approx 16.5\,\mathrm{MeV})$$

Channel	Q (keV)	Notes
$^{16}\mathrm{O}(^{16}\mathrm{O}, p)^{31}\mathrm{P}$	$+7678$	dominant
$^{16}\mathrm{O}(^{16}\mathrm{O}, 2p)^{30}\mathrm{Si}$	$+381$	
$^{16}\mathrm{O}(^{16}\mathrm{O}, \alpha)^{28}\mathrm{Si}$	$+9594$	strongly exothermic
$^{16}\mathrm{O}(^{16}\mathrm{O}, 2\alpha)^{24}\mathrm{Mg}$	-390	endothermic
$^{16}\mathrm{O}(^{16}\mathrm{O}, d)^{30}\mathrm{P}$	-2409	endothermic
$^{16}\mathrm{O}(^{16}\mathrm{O}, n)^{32}\mathrm{S}$	$+1499$	neutron source

And the released deuteron is immediately photodisintegrated: $d + \gamma \rightarrow p + n \quad \Rightarrow$ another neutron supplier.

Oxygen burning: products, neutrons, and timing

- Most abundant nuclei at end of core O burning: $^{16}\mathrm{O}$, $^{24}\mathrm{Mg}$, $^{28}\mathrm{Si}$ (and increasing $^{32}\mathrm{S}$).
- Many channels release neutrons: directly, or via $d + \gamma \to p + n$.
- These neutrons are the seeds for synthesis *beyond* the iron peak — the topic of the next lecture (s-process and r-process).
- Temperature range: $T_9 = 1.5$–2.7 quiescent, 3–4 explosive.

Why neutrons matter now

Above iron, (p, γ) and (α, γ) Coulomb barriers become prohibitive. Heavier elements are built by neutron capture, and the neutron flux first appears here, in oxygen burning.

Time budget

Core O burning lasts only months to a few years in massive stars — vastly shorter than hydrogen burning's $\sim 10^{10}$ y.

Silicon burning — fusion gives up; photodisintegration takes over

- Post-O-burning core: $^{28}\mathrm{Si}$ **and** $^{32}\mathrm{S}$ dominate; $^{28}\mathrm{Si}$ burns first (lower Coulomb barrier).
- But direct fusion is now prohibited:
  - $^{28}\mathrm{Si} + ^{28}\mathrm{Si}$: Coulomb barrier $\approx 38.7\,\mathrm{MeV}$
  - $^{28}\mathrm{Si} + ^{32}\mathrm{S}$: Coulomb barrier $\approx 43.3\,\mathrm{MeV}$

  Both are far above thermal energies even at $T_9 \sim 4$.
- Instead, energetic γ's drive a **photodisintegration rearrangement** — the same mechanism as neon burning, but on a much more extensive scale.

How rearrangement works

$\gamma + ^{A}\mathrm{X} \to ^{A-4}\mathrm{Y} + \alpha$

followed by (α, γ), (p, γ), (n, γ) captures on the surviving nuclei.

Net effect: $^{28}\mathrm{Si}$ is dissolved into α, p, n and rebuilt into heavier nuclei one nucleon at a time.

Time scale

$\tau \sim 10^2$–$10^7\,\mathrm{s}$ — the briefest of all stages.

Quasi-equilibrium and the iron peak

- As T_9 rises through ~ 4, (γ,α), (α,γ), (γ,p), (p,γ), (γ,n), (n,γ) pairs all run at comparable rates $\Rightarrow$ a nuclear statistical quasi-equilibrium forms.
- Abundances drift toward the *most tightly bound* nuclei — the binding-energy curve has its peak at ^{56}**Fe**, ^{56}**Ni**, ^{62}**Ni**.
- Temperatures: $T_9 = 2.8$–4.1 (quiescent), $T_9 = 4$–5 (explosive).

End of the line for fusion

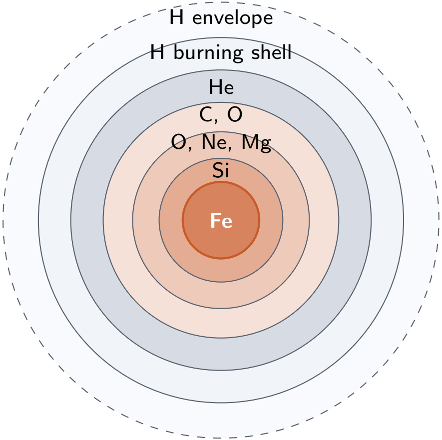
Beyond the iron peak, fusion costs energy — it no longer *powers* the star, it *drains* it. The core is now an iron core on the brink of gravitational collapse.

All four stages at a glance

Stage	T_9 (quiescent)	T_9 (explosive)	Trigger reaction	Main ashes
Carbon	0.6–1.0	1.8–2.5	$^{12}C+^{12}C$	^{16}O, ^{20}Ne, ^{23}Na, ^{24}Mg
Neon	1.2–1.8	2.5–3.0	$^{20}Ne(\gamma,\alpha)^{16}O$	^{16}O, ^{24}Mg, ^{28}Si
Oxygen	1.5–2.7	3.0–4.0	$^{16}O+^{16}O$	^{28}Si, ^{32}S, $^{24}Mg+$ free n
Silicon	2.8–4.1	4.0–5.0	photodisintegration	iron-peak nuclei

- Each stage is shorter and hotter than the last; neutrino losses dominate energy bookkeeping.
- Stages 2 & 4 are driven by **photodisintegration**, stages 1 & 3 by **heavy-ion fusion**.
- Free neutrons emerge during O burning $\Rightarrow$ seeds for s-process nucleosynthesis above iron.

The onion before iron-core collapse



- Layers from outside in trace the star's history in reverse: the youngest fuel is outermost, the most processed material sits in the core.
- Each interface is an active burning shell.
- Mass fractions of any single layer can vary widely, but the *ordering* is universal in massive stars.

What happens next

With an inert **Fe core** growing toward M_{Ch}, photodisintegration of iron robs the core of pressure support $\Rightarrow$ core-collapse supernova.

Summary — advanced burning to the iron peak

- Stars massive enough to clear the Chandrasekhar limit progress through four advanced burning stages: **C** $\rightarrow$ **Ne** $\rightarrow$ **O** $\rightarrow$ **Si**.
- C and O burn by *heavy-ion fusion* through dense forests of compound-nucleus levels.
- Ne and Si burn by *photodisintegration* — the rising T breaks weak bonds ($^{20}\mathrm{Ne}$, then $^{28}\mathrm{Si}$) and the freed nucleons feed (α, γ), (p, γ), (n, γ) chains.
- Free neutrons first appear in O burning, seeding above-iron synthesis (next lecture).
- Si burning ends in nuclear statistical quasi-equilibrium at the **iron peak**: ^{56}Fe, ^{56}Ni, ^{62}Ni — the most tightly bound nuclei.
- Beyond iron, fusion no longer pays $\Rightarrow$ the star is one step from core-collapse supernova.

Next: neutron-capture nucleosynthesis — s- and r-processes.

Thank you

Questions?